

XAVIER ROBERTS-GAAL

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EDUCATION

Harvard University

2022 – present

PhD, Psychology (in progress). AM, Psychology (May 2025). Supervisor: Fiery Cushman ([link](#)).

University of Oxford (New College), Oxford, UK

2017 – 2020

BA (Honours) Psychology & Philosophy, First Class Honours

- Thesis: ‘Two rival pictures of social cognition’. Supervisor: Anita Avramides
- Key classes: How to Build a Brain (Neuroscience + AI), Clinical Social Cognitive Neuroscience
- Activities: Oxford Nightline (crisis hotline), Oxford Union (debating prize), US debate coaching

AWARDS AND HONORS

- National Defense Science and Engineering Graduate Fellowship, US Air Force Office for Scientific Research, >\$200,000, *accepted*
- National Science Foundation Graduate Research Fellowship, >\$150,000, *declined*
- Lightspeed (Jaan Tallinn), \$15,000, awarded for project on dual-process cognition for AI safety
- Norman Henry Anderson Graduate Psychology Fund at Harvard, \$7,500, awarded for cross-cultural research project with the Himba in Kunene Region, Namibia
- Stimson Fund at Harvard, \$1,000, awarded for projects in moral decision-making
- Harvard Mind Brain Behavior Interfaculty Initiative, Graduate Student Award, \$820, for “Compositionality in cognition: the language of thought and its discontents”
- AI Safety Student Team at Harvard, \$300, awarded for project on measuring cognitive capacities in LLM agents
- New College, Oxford Examination Prize for scholarship in Finals
- USA National Merit Scholar Finalist
- NCFL Grand National Tournament, 2x Semifinalist (top 3 high school Lincoln-Douglas debater in the USA at national championship)

PAPERS [*shared first-authorship. Tags: ^m (moral cognition), ^c (culture and social cognition), ^{AI} (AI)]

Representative papers

1. ^c **Roberts-Gaal, X.**, Bolic, M., & Cushman, F. A. (2025). Environmental variability shapes the representational format of cultural learning. *Proceedings of the National Academy of Sciences*. <https://doi.org/10.1073/pnas.250528312>
2. ^{AI, m, c} Voudouris, K*, **Roberts-Gaal, X***, Buhl, M., Irving, G., & Summerfield, C. AI alignment is a human problem. https://osf.io/preprints/psyarxiv/zqngj_v1

3. ^m Le Pargneux, A.*, **Roberts-Gaal, X.***, & Cushman, F. A. Hard bargains and even splits: Fairness judgments track bargaining power across diverse cultures. Under review at *Nature Communications*. https://osf.io/preprints/psyarxiv/3uqks_v2

Working and in prep papers

4. ^{AI} Zhou Hong, S., Kleinman, A., Mathiowetz, A., Howes, A., Cohen, J., Ganta, S., Letizia, A., Liao, D., Pahari, D., **Roberts-Gaal, X.**, Righetti, L., & Torres, J. (2026). Measuring mid-2025 LLM-assistance on novice performance in biology. <https://arxiv.org/abs/2602.16703> [first large-scale experiment of its kind; papers in AI evaluation are often published only on ArXiv]
5. ^m Law, K. F., ..., **Roberts-Gaal, X.**, ..., & Syropoulos, S. (2026). Social discounting outpredicts temporal discounting in intergenerational judgment: A many labs investigation. https://osf.io/6qyv7_v1
6. ^{m, c} **Roberts-Gaal, X.***, Le Pargneux, A.*, Mungunda, M., Tjikuvua, P., Fredrik, C., Hartley, V., Cushman, F. A., Kroupin, I., & Davis, H. Moral judgment and proportional contribution in a small-scale society.
7. ^{AI} **Roberts-Gaal, X.**, Williams, T., Turner, J. M., Mykhailiuk, J., Wisdom, S., Duus, E., Wei, K., & Kolt., N. Managing multi-agent AI risks: Lessons from energy infrastructure and financial markets.
8. ^c **Roberts-Gaal, X.***, Zeng, T. C.*, Mungunda, M., Tjikuvua, P., Fredrik, C., Hartley, V., Henrich, J., Cushman, F. A., Kroupin, I., & Davis, H. Model-free and model-based control beyond WEIRD.
9. ^{AI} Paskov, P.*, Wei, K.*, Hong, S. Z., Bateyko, D., Ezell, C., **Roberts-Gaal, X.**, Guest, E. M., & Bhatt, E. RCTs & human uplift studies: Methodological challenges and practical solutions for frontier AI evaluation. Under review at *AI Ethics and Society* 2026. <https://arxiv.org/abs/2603.11001>

Published papers

10. ^c **Roberts-Gaal, X.**, Callaway, F., & Cushman, F. A. (2026). Ratchet or hatchet? Modeling the cultural evolution of simplified construals. *Proceedings of the Cognitive Science Society*, 48.
11. Woodley, L.*, **Roberts-Gaal, X.***, Calcott, R.*, & Cushman, F. A. (2026). Experimenter demand effects in online psychology experiments. *Open Mind*, 10, 998–1016. <https://doi.org/10.1162/OPMI.a.367>
12. ^{AI} Ezell, C., **Roberts-Gaal, X.**, & Chan, A. (2025). Incident Analysis for AI Agents. *Proceedings of the AAAI ACM Conference on AI, Ethics, and Society*, 8(1), 865–878. <https://doi.org/10.1609/aies.v8i1.36596>
13. ^{AI} Wei, K.*, Paskov, P.*, Dev, S.*, Byun, M.*, Reuel, A., **Roberts-Gaal, X.**, Calcott, R., Coxon, E., & Deshpande, C. (2025). Position: Human baselines in model evaluations need rigor and transparency (with recommendations & reporting checklist). *International Conference on Machine Learning* (spotlight poster). Version of record: <https://github.com/kevinlwei/human-baselines>
14. ^c Levy, A., **Roberts-Gaal, X.**, & Cushman, F. A. (2025). No Evidence for Cost-Benefit Arbitration Between Social Learning Strategies. *Proceedings of the Cognitive Science Society*, 47.

<https://escholarship.org/uc/item/6gs0v9rs>

15. ° **Roberts-Gaal, X.** & Cushman, F. A. (2023). Computational Principles Underlying the Evolution of Cultural Learning Mechanisms. *Proceedings of the Cognitive Science Society*, 45.
<https://escholarship.org/uc/item/78c4w10j>
16. Cuve, H. C., Stojanov, J., **Roberts-Gaal, X.**, Catmur, C., & Bird, G. (2021). Validation of Gazeport Low-Cost Eye-Tracking and Psychophysiology Bundle. *Behavior Research Methods*, 54, 1027–1049. <https://doi.org/10.3758/s13428-021-01654-x>

CONFERENCE PRESENTATIONS (first-author only)

- “Modeling the cultural evolution of task construals.” *Cognitive Science Society*, Rio de Janeiro, Brazil, July 2026 (Poster, accepted)
- “Workshop: The cognitive science of AI alignment” *Cognitive Science Society*, Rio de Janeiro, Brazil, July 2026 (accepted; among the highest-rated workshops).
- “No evidence for experimenter demand effects in three online psychology experiments.” *Society for Philosophy and Psychology*, Johns Hopkins University, June 2026 (Poster)
- “A theory of cultural simplification.” *Society for Philosophy and Psychology*, Johns Hopkins University, June 2026 (Poster)
- “Hard bargains and even splits: Fairness judgments track bargaining power across diverse cultures.” *Human Behavior and Evolution Society*, **finalist for Young Investigator award**, Rabat, Morocco, May 2026 (Talk)
- “Modeling the cultural evolution of task construals.” *Cultural Evolution Society*, Rabat, Morocco, May 2026 (Talk)
- “Third-party moral judgments about negotiations are sensitive to bargaining power.” *Cognitive Science Society*, San Francisco, USA, July 2025 (Poster)
- “Third-party moral judgments about negotiations are sensitive to bargaining power.” *Society for Philosophy and Psychology*, Carnegie Mellon University, June 2025 (Poster)
- “Environmental variability shapes the cognitive engine of culture.” *Society for Philosophy and Psychology*, Purdue University, June 2024 (Poster)
- “Computational principles of cultural learning.” *Cognitive Science Society*, Sydney, Australia, July 2023 (Talk)
- “Uncertainty in moral judgment.” *Society for Philosophy and Psychology*, University of Pittsburgh, June 2023 (Poster)
- “Uncertainty in moral judgment.” *European Experimental Philosophy Conference*, University of Zürich, August 2023 (Poster)

PROFESSIONAL EXPERIENCE

Advisor, confidential clients

Nov 2025 – Present

- *Ad hoc* grantmaking advisory, including portfolio evaluation and impact strategy. Largely in biosecurity and AI resilience.

Prof. Zoë Johnson King**Sep 2023 – May 2024***Research Assistant, Philosophy Department*

- Research assistance, including distilling the relevant psychology/neuroscience literature, for Prof. Johnson King's book, *Praiseworthiness: Its Contours and Limits*

McKinsey & Company, London, UK; Global projects**Nov 2020 – Jun 2022***Business Analyst*

- Helped manage creation and deployment of 4 machine learning models driving £200m revenue impact for top UK bank. Won key stakeholder support. Identified, designed, and implemented 3 marketing initiatives to capture high potential customer segments
- Defined technical architecture, governance, and data ethics approach for cross-govt data linking effort (50+ datasets) joining up social services to support vulnerable individuals
- Created P&L model and identified ~£50m store productivity uplift for UK retailer's strategy
- Designed and modelled incentives to boost efficacy of £200m spend for top-5 auto group
- Conducted 3 due diligences, on packaging, ecommerce/fashion, and translation companies
- Led training sessions for 10k employees in largest Agile transformation in the energy sector
- Consistently ranked in top quintile of performance. Contributed to 3 proposal wins

*Summer Business Analyst***Jun 2019 – Aug 2019**

- Implemented analytics system in top-10 EU corporate bank to improve capital management
- Conducted growth strategy and due diligence for heritage footwear company which later IPO'd

Human Information Processing and Neurotheory Labs, Oxford, UK ([link](#))**Jun 2020 – Nov 2020***Research Assistant – PIs: Chris Summerfield, Andrew Saxe*

- Investigated the learning dynamics of deep neural networks parameterized by Hebbian and top-down feedback components to identify which rules yield qualitative signatures of humanlike learning (e.g., stagelike transitions in loss function, illusory correlations)
- Used formal analysis, representational similarity analysis, and multidimensional scaling, and developed pipelines in Python (JAX, TensorFlow)
- Scaled up experiment using texture synthesis patterns and naturalistic computer graphics
- Worked on project implementing meta-learning of intrinsic reward to model mood disorders
- Contributed to journal club and lab activities

Bird Lab, Oxford, UK ([link](#))**Apr 2019 – Jun 2020***Research Assistant – PI: Geoff Bird*

- Co-authored a new validation protocol for low-cost eye-trackers used in psychophysics (Gazepoint GP3HD and biometric package). Identified ways to minimize data loss
- Gathered data for experiments on interoception, emotional gaze in autism, and empathy
- Built an R-Shiny web app to process and analyze data, improving the lab's workflow efficiency

Moonshot Global Consulting, Washington, DC, USA; Global projects**Jan 2018 – Jun 2019***Intern*

- Supported on monitoring and evaluation projects for Australian, USA, and UK development agencies, incl. AUD 15 million project growing the APAC impact investing ecosystem

- \$400,000 World Bank project implementing a mechanism for private sector feedback on public services, and \$45 million, 4.5-year open innovation project funded by USAID, DfID, Sida, and Omidyar Network

MENTORSHIP

- Mira Becker (Senior Honors Thesis, received Psychology Faculty Prize; now at More in Common)
- Nikola Jurkovic (Tutor for Special Concentration in AI and Society; now at Model Evaluation and Threat Research, the premier AI evaluations nonprofit)
- Marija Bolic (UOttawa, summer internship; now PhD student at University of Toronto)

SERVICE

- *Ad hoc* peer review for *Nature Communications*, *Journal of Personality and Social Psychology*, *Evolution & Human Behavior*, *Synthese*, *Cognitive Science Society*, *AI Ethics and Society*, *NeurIPS MP2 Workshop*
- Facilitator, **AI Safety Student Team**, AI safety fundamentals for Harvard/MIT undergrads
- Mentor, **PPREP** (supporting PhD/RA applicants from underrepresented groups)
- Convened interdisciplinary PhD reading group on **compositionality** in cognition, Spring 2023
- Created professional PowerPoint templates for lab

SKILLS

<i>Analytical</i>	Statistical modeling (incl. Bayesian methods), deep learning, computational cognitive models, evolutionary models, game theory, RL, LLMs
<i>Languages (programming)</i>	Python (PyTorch, TensorFlow, JAX, NumPy, Pandas, Scikit-Learn, PsychoPy, PyGaze), R, Julia, intermediate MATLAB
<i>Software</i>	Tableau, SPSS, JASP, Alteryx, Word, Excel, PowerPoint, Qualtrics, Gorilla, Git, Atlassian tools (e.g., Jira, Confluence, Trello, Bitbucket), Azure DevOps
<i>Hardware</i>	Gazepoint GP3HD eye-tracker and biometrics bundle, Biopac
<i>Languages (human)</i>	Native English, intermediate Mandarin
<i>Interpersonal</i>	Leadership, project management, empathic/crisis listening, senior executive communication, coaching, workshop facilitation, and interviewing

ADDITIONAL INFORMATION

Citizenship	USA, Australia, and Hungary
Interests	Rock climbing, rowing, travel, swimming, poetry, piano, tea, reading, and thinking a lot about minds

Volunteering Co-organizer, UK AI Security Institute Alignment conference; advised [Taimaka](#), a Nigeria-based hunger/poverty charity (recipient of D-Prize, funded by USAID); mentored neuroethics debates for high schoolers